

Stock Project

Team name: Stock Project | Team member: Scottie Dunnavaan | Deliverable: Project Proposal Update — Dataset Description (PS3)

1. Problem Statement

Retail traders compete against institutions that run algorithmic and high-frequency strategies on proprietary data and infrastructure. Manual chart reading cannot keep pace with the volume of live order-book updates and is prone to emotional bias. This project targets one narrow, measurable version of that gap: **short-horizon (1-5 minute, intraday) price-direction prediction**. Given recent limit order book (LOB) and price-history data for a stock, a model estimates the probability that the mid-price will move *Up*, *Neutral*, or *Down*, and those probabilities are turned into entry/exit signals under strict risk limits.

Why it matters: whether deep learning can extract a usable signal from data an individual can actually access, and how much of the published LOB results survive outside curated benchmarks, are open and practical questions. Because financial data is noisy and non-stationary, the project emphasizes rigorous, leakage-free evaluation over headline accuracy; a well-evaluated negative or marginal result is still a valid outcome.

2. Motivation & Objectives

Motivation. I have a long-standing interest in the mathematics of markets, inspired by quantitative firms that trade systematically using mathematical and statistical-learning methods. This is a personal, hobby and portfolio project: I want to learn the foundations of algorithmic trading and deep learning by building an end-to-end system (paper trading only in this course). **Why deep learning:** LOB data is high-dimensional (prices and volumes across many levels), sequential, and noisy. Hand-crafted indicators discard much of that structure, while convolutional and recurrent/attention models can learn spatial and temporal patterns directly. DeepLOB [1] showed this on benchmark data; this project tests how far it carries to live U.S. equity data.

Objectives (task type: 3-class classification of price direction):

- **O1 (core):** Reproduce DeepLOB on the FI-2010 benchmark to validate the pipeline.
- **O2 (core):** Build classical baselines (Random Forest, XGBoost) on technical and order-book features.
- **O3 (core):** Train deep models on self-recorded U.S. equity data for 1-5 minute direction.
- **O4 (core):** Evaluate with chronological splits and transaction costs.
- **O5 (stretch):** Fuse micro (LOB) and macro (price-history) models, add a risk-managed decision layer, and run paper trading.

3. Dataset Description

The data plan has three stages: a public benchmark (FI-2010) to validate the pipeline, a purchased historical pull (Databento) to bridge the gap while recording begins, and self-recorded live U.S. equity data for the primary experiments, supported by free daily data.

Phase 1 — FI-2010 benchmark (validation). 5 Nasdaq Nordic stocks, 10 trading days (June 2010), ~4 million order-book events reduced to ~396,000 normalized time steps; top-10 bid/ask price and volume levels; Up/Stationary/Down labels at five prediction horizons. Free and publicly available [2], distributed as plain text files (~1.74 GB across normalization variants). Each file is organized feature-rows × time-step-columns: rows 1-40 are basic bid/ask price and size for the top 10 levels per side, rows 41-144 are additional handcrafted features from the original benchmark study, and rows 145-149 are the five-horizon labels. Time steps are event-indexed (after every 10 order-book events) rather than wall-clock timestamped. Used exclusively to reproduce DeepLOB and validate the pipeline before training on live data; results are not assumed to transfer to live U.S. small-cap equities.

Phase 1.5 — Databento historical pull (bridging). To begin training on domain-matched U.S. small-cap data immediately, rather than waiting for Phase 2 to accumulate, a historical data credit from Databento is used to pull Level 1 and Level 2 (Nasdaq TotalView-ITCH) history for a screened small-cap universe. This supplements, and is gradually superseded by, the self-recorded dataset below.

Phase 2 — self-recorded U.S. equity data (primary, ongoing). A custom asyncio-based Python ingestion engine (market_data_engine) streams live market data directly from three data sources, selected after evaluating cost, exchange coverage, and programmatic (non-display) API access across several vendors:

Source	Contents	Cost / constraints	Role
Moomoo OpenAPI	NYSE ArcaBook order-book depth (Level 2)	Free with a minimum account balance; subject to change	Primary L2 depth, exchange 1 of 2
Interactive Brokers (IBKR) TWS API	Nasdaq TotalView order-book depth (Level 2)	Paid market-data subscription	Primary L2 depth, exchange 2 of 2
tastytrade Open API	Real-time, fully consolidated Level 1 quotes and trades	Free	Consolidated volume/quote context for every candidate
Yahoo Finance (yfinance)	Daily OHLCV history	Free; rate-limited	Macro market-regime features, baselines

Target universe. Small-to-mid-cap U.S. equities (approximate market capitalization \$1M–\$500M), selected daily rather than from a fixed list. A scanning and tiering process identifies candidates from top gainer/loser lists, filters by market capitalization and recent news, and escalates a ticker from basic snapshot monitoring to continuous Level 1 and then Level 2 streaming as its price move, relative volume, and order-book density cross defined thresholds — so the recorded dataset captures many different tickers' behavior rather than a handful of repeatedly-sampled names.

Format. Unlike FI-2010's fixed text matrices, streamed quotes, trades, and order-book snapshots are serialized as tabular Parquet files (one row per update), partitioned by date, session, and symbol, stored locally and synced to cloud storage for durability. L1 rows carry a UTC timestamp, symbol, session, source, bid/ask price and size, last trade price and size, and cumulative volume; L2 rows carry the same identifying columns plus price/size pairs for each tracked depth level on both sides — the live equivalent of FI-2010's 40 basic features, but with real wall-clock UTC timestamps rather than event-count indices.

Status. The ingestion engine is under active development. An initial implementation surfaced data-integrity issues in its buffered file-writing logic, which are being corrected before continuous daily recording begins. FI-2010 and free daily bars support model development in the meantime.

Preprocessing & feature engineering:

- **Input shapes:** sliding matrices of recent LOB snapshots (100 time steps × 40 features: 10 bid/ask price and volume levels per side), paired with 1-minute OHLCV sequence windows.
- **Normalization:** rolling z-score normalization computed using only past data to prevent lookahead bias; log returns for stationarity.
- **Labeling:** Up/Neutral/Down targets from the thresholded percent change in smoothed future mid-price over a 1–5 minute horizon. Thresholds are fit on training splits for class balance; remaining imbalance is handled with weighted cross-entropy loss.
- **Splits:** strictly chronological train/validation/test, with explicit temporal gaps between splits so overlapping sliding windows cannot leak information.

References

- [1] Zhang, Z., Zohren, S., & Roberts, S. (2019). DeepLOB: Deep Convolutional Neural Networks for Limit Order Books. *IEEE Transactions on Signal Processing*, 67(11), 3001–3012.
- [2] Ntakaris, A., Magris, M., Kannianen, J., Gabbouj, M., & Iosifidis, A. (2018). Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*. Dataset: etsin.fairdata.fi/dataset/73eb48d7-4dbc-4a10-a52a-da745b47a649